

Problem 1

用线性回归拟合曲线 (例2)

编号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
x	2	5	7	10	14	19	26	31	34	38	45	52	53	60	65
y	54	50	45	37	35	25	20	16	18	13	8	11	8	4	6

X: 病人住院天数, Y 病人预后指数, 选择合适曲线拟合 $Y=F(X)$ 并评估其拟合优度。

Problem2: Carry out the multi-variate linear regression on the following data:

$Y=aX_1+bX_2+cX_3+dX_4+e$, and find the Adjusted R-square

表15-2 27名糖尿病人的血糖及有关变量的测量结果

序号 i	总胆固醇 (mmol/L)	甘油三脂 (mmol/L)	胰岛素 (μ U/ml)	糖化血 红蛋白(%)	血糖 (mmol/L)
	X_1	X_2	X_3	X_4	Y
1	5.68	1.90	4.53	8.2	11.2
2	3.79	1.64	7.32	6.9	8.8
3	6.02	3.56	6.95	10.8	12.3
4	4.85	1.07	5.88	8.3	11.6
5	4.60	2.32	4.05	7.5	13.4
6	6.05	0.64	1.42	13.6	18.3
7	4.90	8.50	12.60	8.5	11.1
8	7.08	3.00	6.75	11.5	12.1
9	3.85	2.11	16.28	7.9	9.6
10	4.65	0.63	6.59	7.1	8.4
11	4.59	1.97	3.61	8.7	9.3
12	4.29	1.97	6.61	7.8	10.6
13	7.97	1.93	7.57	9.9	8.4
14	6.19	1.18	1.42	6.9	9.6
15	6.13	2.06	10.35	10.5	10.9
16	5.71	1.78	8.53	8.0	10.1
17	6.40	2.40	4.53	10.3	14.8
18	6.06	3.67	12.79	7.1	9.1
19	5.09	1.03	2.53	8.9	10.8
20	6.13	1.71	5.28	9.9	10.2
21	5.78	3.36	2.96	8.0	13.6
22	5.43	1.13	4.31	11.3	14.9

Problem3

2. (1) Use the forward-difference formulas and backward-difference formulas to determine each missing entry in the following table.

x	$f(x)$	$f'(x)$
0.0	0.000 00	
0.2	0.741 40	
0.4	1.371 8	

(2) Data in part(1) were taken from the function $y = e^x - 2x^2 + 3x - 1$. Compute the actual errors in part(1), and find error bounds using the error formulas.